



Fluid circulation and deformational gradient in north-Pyrenean flyschs: Example from the Saint-Jean-de-Luz basin (France)

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ABSTRACT

The relationships between fluid circulation and deformation are one of the issues of the sedimentary basin study. In the Cretaceous flysch of the Saint-Jean-de-Luz basin, the evolution of folds geometry and the increasing volume of calcite-filled fractures and veins evidence a northward deformational gradient along the French Basque coast. A combined approach is proposed to assess the corresponding physical and chemical conditions: the microthermometric study of fluid inclusions in calcite sampled in different generations of veins and fractures and the X-ray diffraction analysis of clay minerals from adjacent marl layers. Salinity of the trapped H₂O–CaCl₂–NaCl fluids increases with depth in the series, in good agreement with salinity gradients reported in sedimentary basins. Dispersion of the data also increases with depth from 0.3 to 1.3 wt.% NaCl eq. in the shallowest formation (Haizabia flysch) to 9.1 to 23.0 wt.% NaCl eq. in the deepest formation (Guéthary flysch). Minimal trapping temperatures of the fluids in the Haizabia and Socoa flyschs (79 and 102 °C, respectively) are consistent with the temperatures estimated from the depth of burial, which did not exceed 5 km, in good agreement with the stability of the smectite–illite–kaolinite assemblage found in marls. In addition, the kaolinite proportion significantly decreases with depth in the series, as a potential consequence of climate changes and diagenetic transformations, whereas the increasing dispersion of illite crystallinity data might indicate fluid–mineral interactions. We propose a synthetic model of fluid circulation in the folded series that involves the mixing of mainly horizontal fluid circulation (potentially meteoric) with an upward flow of high-salinity fluid throughout the deepest formations (potentially related to underlying evaporite-rich layers). The northward deformational gradient, as exposed along the French Basque coast, is likely to be responsible for such a vertical circulation, by increasing the volume of fracture (particularly cross-cutting fractures) in the deeply buried formations.

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1. Introduction

The role of fluids in fracturing and deformation of sedimentary rocks has been extensively studied for a long time (e.g. Secor, 1965; Sibson, 1994). Fluids composition and physical conditions are heterogeneous within deforming series (Fischer et al., 2009) and vary over time, before, during and after folding (e.g. Evans, 2010; Roure et al., 2005). Several studies have implemented fluid inclusions microthermometry to structural analysis in order to constrain those changes in fluid conditions during folding and fracturing of sedimentary rocks (Evans et al., 2012, and references therein).

Chemistry of fluids involved in sedimentary basins is still an issue as a wide range of various and remote mechanisms are involved (e.g. Bjørlykke, 1993, 1994; Hanor, 1994). Composition of the fluid inclusions hosted in diagenetic minerals results from the initial composition of fluids trapped during deposition, from diagenetic exchanges with

sediments, and from mass transfers at basin scale (Anastasio et al., 2004; Hanor, 1994; Vilasi et al., 2009). Processes such as infiltration of brines (Hanor, 1994; Sanford and Wood, 1991) or dissolution of high-chloride evaporites (Macpherson, 1989) can also be involved in the production of fluids saltier than seawater (fluid salinity data referenced by Hanor, 1979 and Ranganathan and Hanor, 1988). Moreover, migration of fluids requires thermal or physical driving forces such as topographic gradients, sediment compaction, phase changes, density gradients and tectonics (Deming, 1994) and sufficient porosity and permeability of the anisotropic sedimentary rocks, which are controlled by diagenesis and fracturing (e.g. Marrett and Laubach, 1997; Nelson, 2001).

The Cretaceous flysch of the Saint-Jean-de-Luz basin, France (Fig. 1) provides a clear example of kinematic relationships between fracturing and folding (Bodou, 1972; Robert, 1979) in series where the sedimentary processes, the stratigraphy and the tectonic background are fairly well documented (Mathey, 1986; Mulder et al., 2009; Razin, 1989). The deformation kinematics has been established thanks to very good exposure of the folded structures on the French Basque coast, but the physical conditions associated with the folding process and the fractures and veins formation are still poorly documented.

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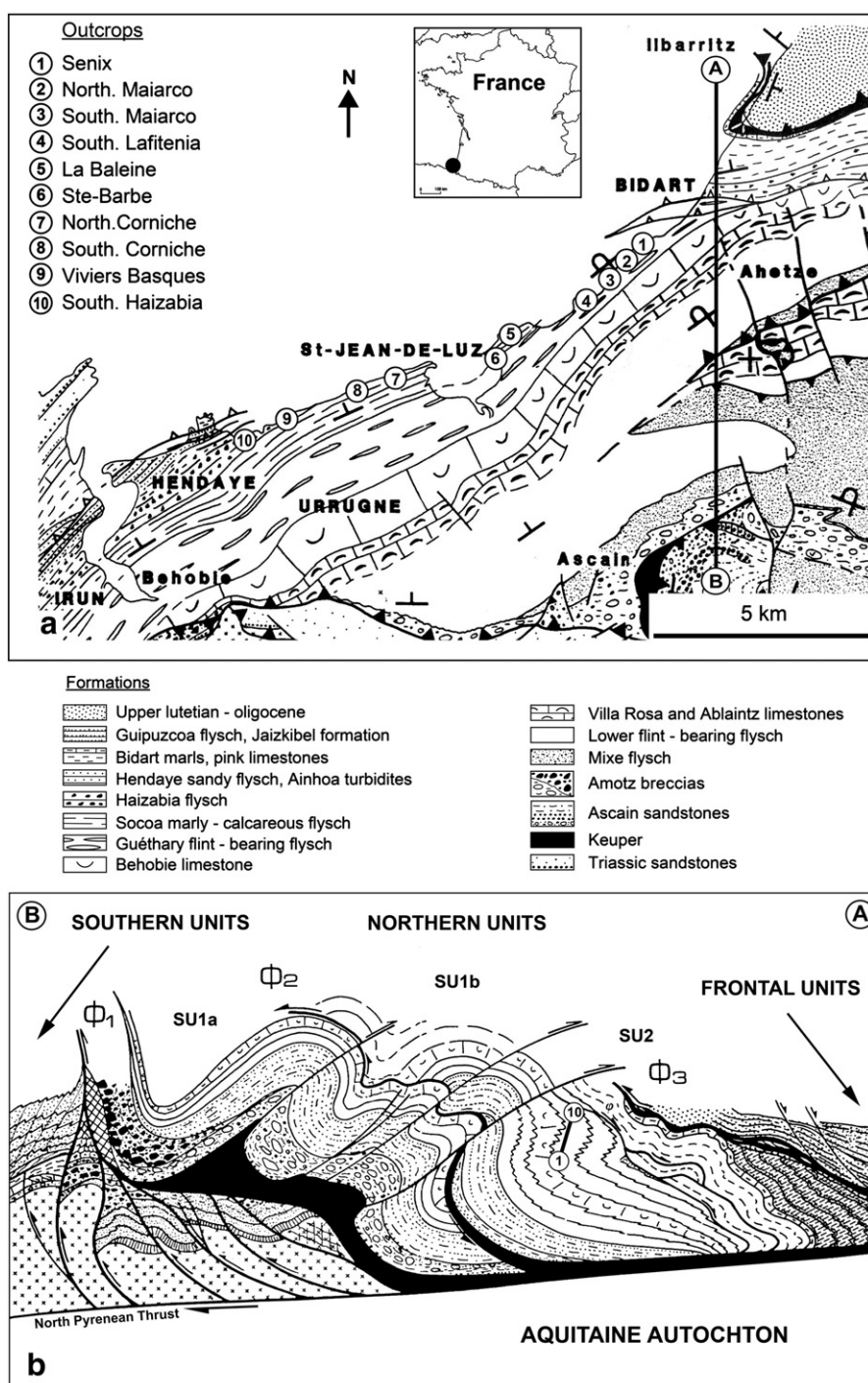


Fig. 1. Structure of the Basque Pyrenees. Simplified map (a) and cross-section (b) of the geological formations on the French Basque coast. Positions of the sampled outcrops (1 to 10) are projected into the cross-section. Modified after Razin, 1989.

The present study consists of a preliminary investigation of the fluid–mineral interactions in the Saint-Jean-de-Luz basin and the northern Pyrenees. It aims to establish the relationships between the observed northward deformational gradient, the burial of the basin and local tectonic causes, and to provide a synthetic model of fluid circulation constrained by the sedimentary and structural data at basin scale. The outcrops of the French Basque coast were chosen as they provide greatly exposed structures and abundant mineralization of venous and fibrous calcite. Their fluid inclusions content was investigated by microthermometry to characterize the fluid composition and their trapping conditions. Besides, the marl layers available throughout the flysch

series were sampled to analyse the clay mineralogy and illite crystallinity by X-ray diffraction.

2. Geological setting

The alpine evolution of the Pyrenees is characterized by the displacement of the Iberian microplate relative to the European plate during the Cretaceous. Rifting and oceanic accretion in the Bay of Biscay have led to the counterclockwise rotation and the southward displacement of Iberia with respect to Europe (Olivet, 1996; Sibuet and Collette, 1991; Srivastava et al., 2000). The north-Pyrenean flyschs deposited at that

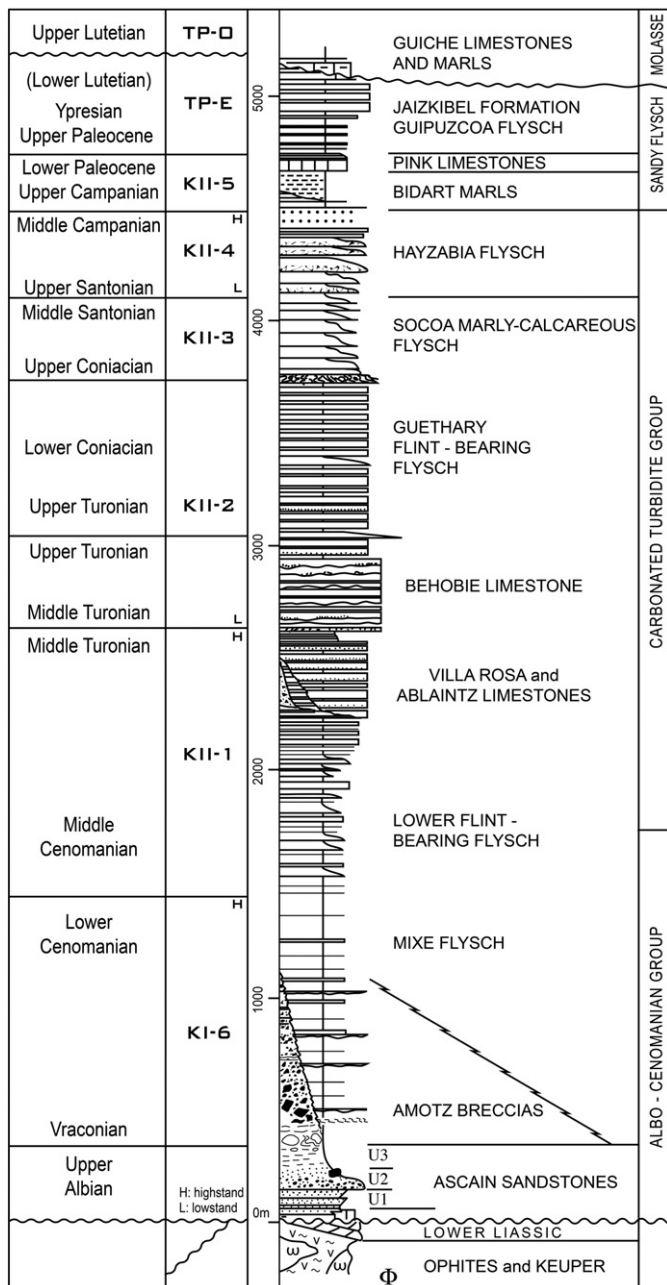


Fig. 2. Regional stratigraphic column. The three studied formations in the Saint-Jean-de-Luz basin are part of the Carbonated turbidite group, and constitute a 1400-to-1800 m-thick pile. Modified after Razin, 1989.

time were deformed by the convergence during Cretaceous to Oligocene times.

The three formations studied in the Saint-Jean-de-Luz basin are exposed along the coastline between Bidart and Hendaye (Figs. 1 & 2) and consist in the Carbonated Turbidite Group (Razin, 1989). As described by Feuillée and Sigal (1965) and Mathey and Sigal (1976), the 600-to-800 m-thick Guéthary flint-bearing flysch¹ (lower Coniacian) is made up of flint-bearing calcarenites and calciltites interlayered with ten-centimetres-thick marly layers. The 350-to-400 m-thick Socoa marly-calcareous flysch¹ (upper Coniacian-middle Santonian) has a higher amount of marl as a consequence of a greater distance from the source

due to a deepened system and of a change in the sediment supply from the platform. The 390 m-thick Haizabia flysch (upper Santonian-middle Campanian) has a higher proportion of coarse-grained layers due to the basin filling during the Campanian compressive tectonic regime (Mulder et al., 2009). On the basis of the stratigraphical (Fig. 2) and structural data of Razin (1989), the minimal depth of burial was assessed at 1.3 to 2.0 km at the top and 1.8 to 2.5 km at the bottom of the Haizabia flysch, 2.3 to 3.0 km at the bottom of the Socoa flysch, and 3.1 to 4.1 km at the bottom of the Guéthary flysch.

3. Deformation kinematics and veins and fractures description

The geological formations of the Saint-Jean-de-Luz basin are structurally involved in the Northern units of the Basque Pyrenees (Razin, 1989). They consist of Mesozoic displaced cover whose global shear is outlined by the north-westward dipping structures, i.e. fold axial surfaces dipping south-east. In the study area, the deformation increases northwards from a homocline area (Urrugne unit, SU1) – upright to at most gently inclined folds – to a strongly folded area (Sainte-Barbe unit, SU1b) involving inclined to recumbent folds (Fig. 3.6). The dip of axial plane progressively decreases from the southern to the northern outcrops, where recumbent folds are observed in the Guéthary flysch (Fig. 4a). The interlimb angle also decreases, reflecting an increasing tightness of the folds (Fig. 4b), from open folds between the Haizabia (Fig. 1, outcrop 10) and the Sainte-Barbe locality (outcrop 6) to tight folds in the Guéthary area (outcrop 4 to 1). This northward evolution of the axial planes and of the tightness of folds evidences an increase of strain towards the lower part of the series. Robert (1979) and Razin (1989) concluded to a north-west shearing strain. In such a context, localized blockages of the interlayer sliding process (Robert, 1979), which is widespread in the series, can be regarded as one of the main causes of the folding.

Fracturing occurred both during and after the folding episode and therefore results in three main groups of fractures and veins: early-folding, syn-folding and late fractures and veins (Fig. 5). Early slight extension along the fold axis direction has led to longitudinal fractures, locally wrapped and crossed cut by extrados fractures (Bodou, 1972). Diagonal fractures are the relays of echelon faults and calcite dominoes (T/P associated Riedel shears, Gamond, 1983) formed before the folding episode or during its earliest stages, as suggested by their orientation on the Sainte-Barbe fold (Fig. 1, outcrop 5, and Fig. 3.4). On both the horizontal and vertical limbs of this fold, numerous right lateral shear fractures are observed, but only few conjugate left lateral shear fractures (Fig. 6a). When the layers are unfolded, it appears that all the fractures belong to a same set, which is perpendicular to the bedding, with a compressive direction perpendicular to the fold axis (Fig. 6b). It also appears that the fractures restored from the almost horizontal limb have westward dipping striae whereas the fractures restored from the vertical limb have eastward dipping striae. Therefore, the striae are not in the same direction, neither in the folded nor in the restored unfolded geometry. Such oblique position of the striae (relative to the bedding and to the position in the fold hinge) evidences a gentle to open fold at this early fracturing time. Extrados fractures are the most frequent syn-folding fractures; most are in radial position but some of them have migrated on the vertical limb of inclined folds during hinge migration (Robert, 1979).

With the increasing deformation (tightening of folds and increasing shear), the folding of the thick layers has led to the centimetric to metric dislocation of the hinges (Figs. 3.5 & 5, arrow 3), as observed during the development of chevron folds (Ramsay, 1974). This was accommodated by both wadding of the incompetent layers in the hinges (micro-reverse faults within the hinges, Fig. 5c) and stretching of the reverse limbs (boudinage of competent strata and micro-normal faults, Figs. 3.2 & 5d). Later transverse evolution of the dissociated fold sections has formed either non-cylindrical folds geometry or strike-slip faults that cross-cut the fold hinges (Fig. 5e). Where the folds are strongly tight,

¹ The formations are thereafter abbreviated Guéthary, Socoa and Haizabia flyschs, respectively.

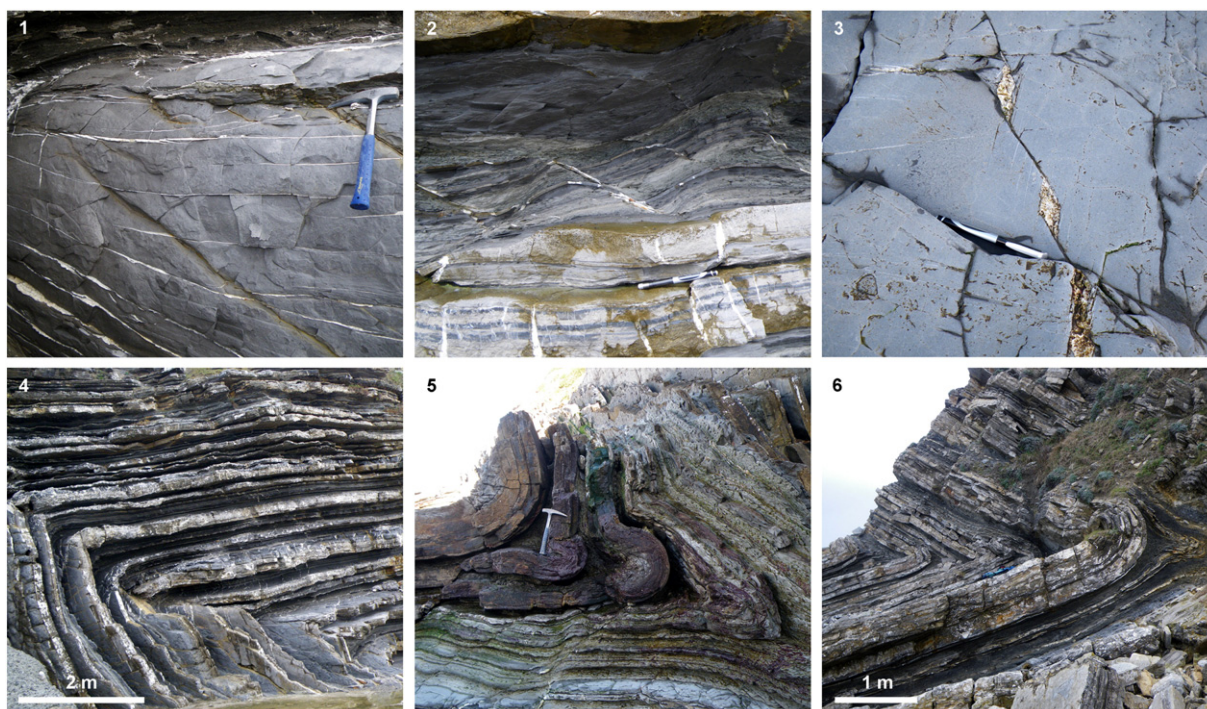


Fig. 3. Photographs of characteristic structures. (1) Extrados fractures on a fold in the Socoa flysch. (2) Stretching of reversed limb in the Guéthary flysch; note the fractures in competent strata and the micro-normal faults in marls. (3) Dextral relay of Riedel P and tension cracks in the Socoa flysch. (4) Overturned fold in the Socoa flysch at the Sainte-Barbe locality. (5) Dislocation of hinge in a dip-right fold in the Socoa flysch. (6) Recumbent fold in the Guéthary flysch; note the hinge wadding on the right-hand.

vertical veins cross-cut both limbs of the isoclinal folds (e.g. outcrop 1, Fig. 7). These veins, more or less parallel to the fold axis, are regarded as late fractures, occurring only in the isoclinal folds of the Guéthary

flysch. The overall organization of the fractures displays, as the fold geometry, an evolution from south to north; the deepest part of the series exposed in the northern area being the most strongly deformed.

Thirty six calcite samples were collected from 10 outcrops along the French Basque coast (Fig. 1) in order to investigate the possible relationships between formation of the structures and nature of the fluids trapped in calcite crystals. In the Haizabia flysch, only the early-folding fractures could have been sampled whereas only the late-fractures were clearly identifiable in the Guéthary flysch. The three generations of fractures were successfully sampled in the Socoa flysch. Twenty marl samples were collected nearby the calcite samples, within the most distal members of the series, avoiding the first ten centimetres exposed to air and seawater alteration.

4. Fluid inclusions microthermometry

4.1. Methods

Double polished sections (~200 to 300 μm thick) of calcite material were prepared following Goldstein's (2003) recommendations and used for microthermometric measurements at the Géosciences Environnement Toulouse (GET) laboratory on a Linkam THMSG-600 heating-freezing stage connected to a programmable thermal control unit. Pure CO_2 fluid inclusions in quartz from Camperio, Central Alps (Mullis et al., 1994; Poty and Stalder, 1974) were used for calibration at CO_2 triple point (-56.6°C) and critical point (31.1°C) temperatures, and synthetic H_2O fluid inclusions in quartz for calibration at H_2O triple point (0.2°C) and critical point (375.13°C) temperatures.

Fluid inclusion microthermometry has been less documented for calcite than for minerals such as quartz (e.g. Diamond, 2003; Roedder, 1984). A specific methodology is required as calcite is a relatively soft and easily cleavable mineral (Goldstein and Reynolds, 1994). The risk of decrepitation of fluid inclusions hosted in common minerals is usually reduced by performing low-temperature measurements first (Bodnar, 2003a), but stretching of the fluids trapped in soft mineral such as sphalerite (Lawler and Crawford, 1983) and calcite (Meunier,

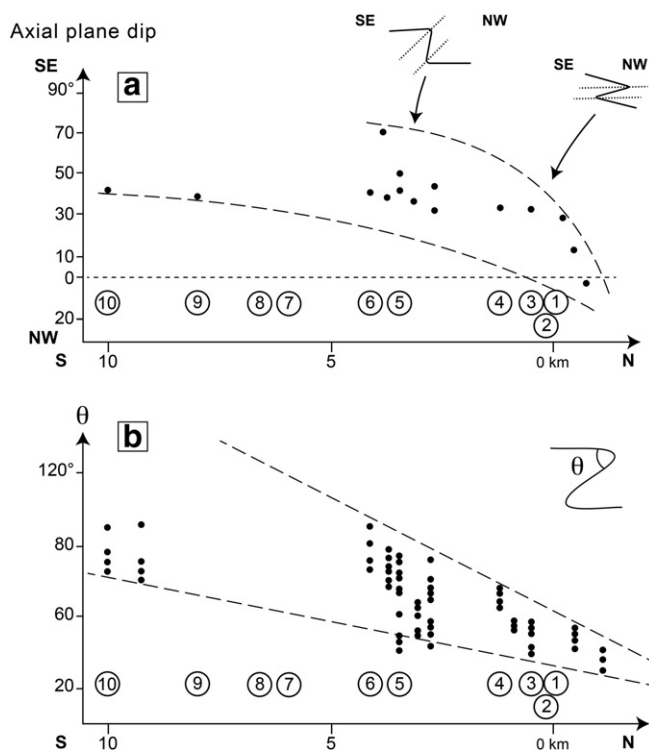


Fig. 4. Evolution of folds shape along the French Basque Coast. Measurement of axial plane dip (a) and interlimb angle (θ) of folds (b), both decreasing northwards. Tightness of the folds increases northwards, towards the bottom of the series. The sampled outcrops are numbered as in Fig. 1. Modified after Robert, 1979.

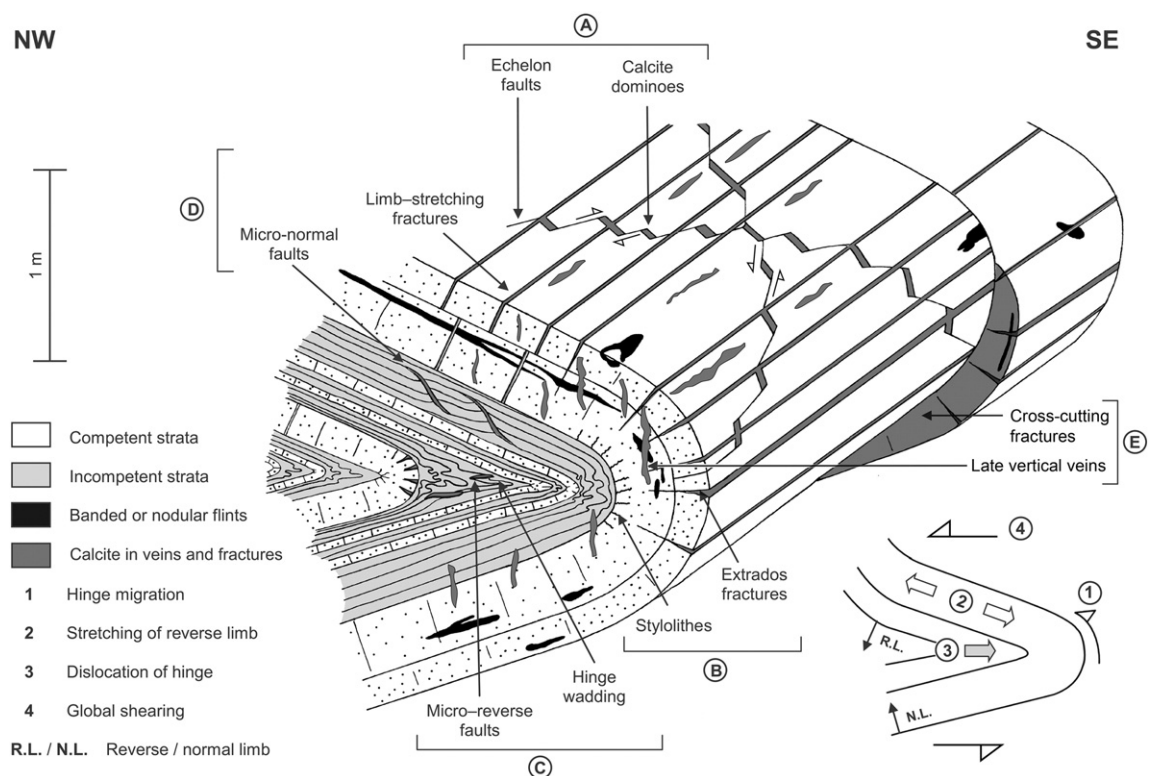


Fig. 5. Synthetic diagram of observed structures. North-westward recumbent synclinal fold displaying: early-folding fractures (A); syn-folding fractures such as hinge features, for competent (B), and incompetent strata (C); fractures in stretched reverse limb (D); and late cross cutting strike-slip fractures and vertical veins (E).

1989) is likely to occur during the freezing stage. Therefore, the high-temperature measurement was conducted first and each sample piece was investigated for one stage only. Homogenization temperature (T_h), first melting temperature (T_e) and final ice-melting temperature (T_m) were measured (Table 1), following the identification criteria of Bodnar (2003b). As recommended by Prezbindowski and Larese (1987), the homogenization temperatures were never exceeded and the heating-cooling rates were kept below 2 °C/min. Accuracy of the measures is estimated at ± 0.2 °C for T_m and ± 1.0 °C for T_h , provided that the vapour bubble volume remained constant. Since metastability phenomena are likely to occur during the heating stage of the low-temperature measurement, the apparition of first melt might have been noticed lately, especially for low-salinity fluids (Goldstein and Reynolds, 1994). As a result, T_e values might have been overestimated and the T_e data are expected to have a lower accuracy.

Raman analyses were conducted at the G2R² laboratory using a Dilor–Labram-type Raman spectrometer to characterize the composition of the vapour phase of 20 selected fluid inclusions, following the instrument calibrations and measurement conditions described by Burke (2001).

4.2. Fluid inclusions description

Low transparency of most of the calcite samples, small number of workable fluid inclusions and poor textural references have restrained the use of the fluid inclusion assemblage concept (FIA), which is recommended in such a sedimentary context (Goldstein, 2001). Instead, the classification of the 115 selected fluid inclusions was restricted to their position relative to cleavage planes, resulting in two batches of fluid inclusions (primary and pseudo-secondary). From the petrographic

observations, and according to the absence of distinguished trends between those two groups of fluid inclusions, we have made the assumption that they were all related to fluid events that are chronologically close (Bodnar, 2003b). The primary and pseudo-secondary fluid inclusions are on average $25 \times 10 \mu\text{m}$ and $7 \times 3.5 \mu\text{m}$ sized, respectively. Two-thirds of the fluid inclusions are compact, with length/width ratio ($l/w < 2$), one-quarter is elongated (l/w up to 5) and a few are very elongated (l/w up to 25).

Reequilibrated fluid inclusions, i.e. whose volume and/or composition have changed after trapping, were recognized from petrographic clues (Fig. 8) such as morphology, size, and liquid/vapour ratios (Bodnar, 2003c). In the Guéthary flysch samples, 57% of fluid inclusions seem to have suffered reequilibration (three-quarters by necking-down) whereas only 28% are questionable among the samples from the two other formations (two-thirds by stretching). Considering that reequilibration does not impact the composition and the density of the trapped fluids in the same way (as discussed later in this article), the reequilibrated fluid inclusions were eliminated for the T_h measurement and distinguished from the intact ones in the T_e and T_m data.

4.3. Composition of the trapped fluids

Estimated T_e values range from -54.6 to -19.9 °C (Fig. 9) and allow to constrain the fluid composition to the $\text{H}_2\text{O}-\text{CaCl}_2-\text{NaCl}$ ternary system as its related eutectic temperature is -55.0 °C (Shepherd et al., 1985). T_m values range from -20.9 to -0.2 °C, which corresponds to salinities ranging from 0.3 to 23.0 wt.% NaCl equivalent (Bodnar, 2003a). Salinity of the trapped fluids increases upwards the series along with dispersion of the data (Fig. 10). The Haizabia flysch samples yield the lowest salinities (0.3–3.6 wt.% NaCl eq.) and the related data are the least dispersed, whereas the Guéthary flysch samples yield the highest and most dispersed values (9.1–23.0 wt.% NaCl eq.). Among the Socoa flysch data (0.3 to 11.8 wt.% NaCl eq.), the fluid trapped in the calcite from syn-folding fractures has the highest salinity (4.6 wt.% NaCl eq.) whereas the pre-folding and late fracture fluids provide

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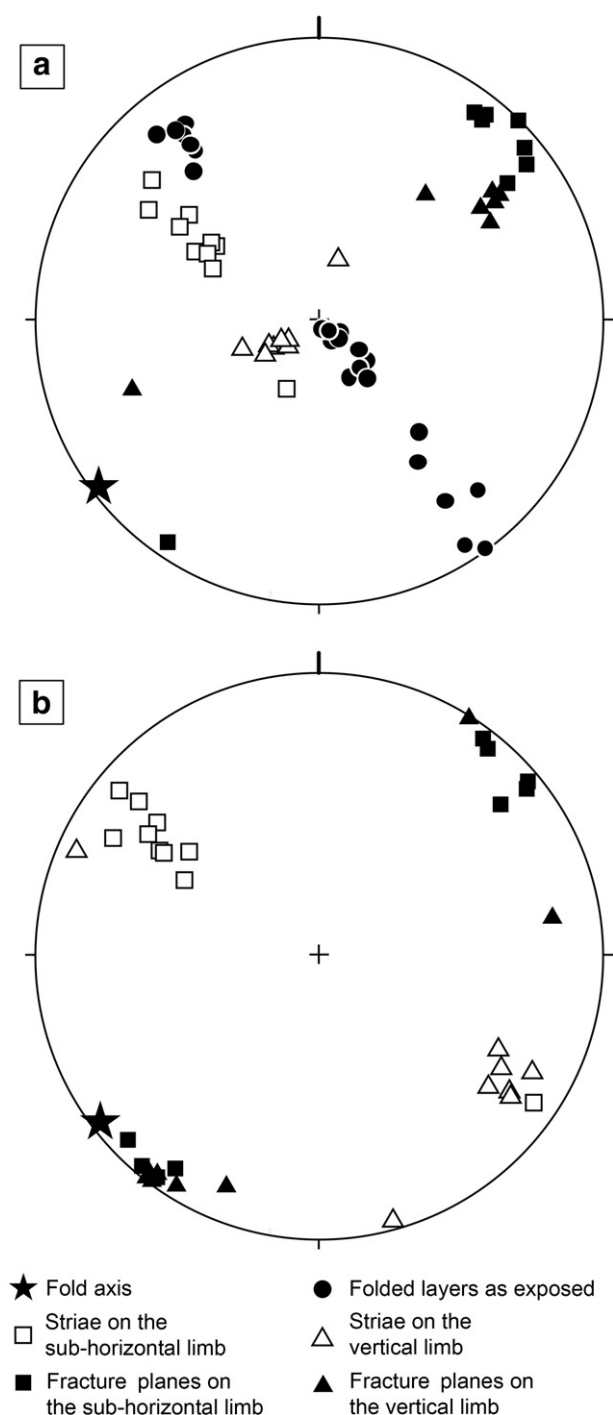


Fig. 6. Stereo projection of measures of the Sainte-Barbe fold. Most of the fractures belong to the same right lateral shear set; orientation of the striae is not consistent with a single direction neither before (a) nor after (b) the folded position is restored, but instead when the interlimb angle is considered at a gentle fold stage. Equal angle projection of the data has been performed with Stereonet software (Allmendinger et al., 2012).

slightly lower salinities (3.6 and 1.4 wt.% NaCl eq., respectively). This difference is also observed in the Guéthary flysch (13.8 and 9.4 wt.% NaCl eq. for syn-folding and late fracture samples, respectively). There is no specific trend for primary and pseudo-secondary fluid inclusions; likewise, no shift was noticed between intact and reequilibrated fluid inclusions (as discussed later).

On the basis of the Raman spectroscopy data, it appears that the vapour phase of the selected fluid inclusions mainly consists of water vapour. Methane was found in trace amounts in the samples from the

Socóa flysch and in higher amounts, yet not measurable, in the samples from the Guéthary flysch (Fig. 10).

4.4. Trapping conditions

The fluid inclusions selected for the high-temperature measurement provide T_h values ranging from 41 to 151 °C. The data from the Haizabía and Socóa flyschs are distributed in unimodal histograms (Fig. 11a & b) centred on their mean values (78 and 102 °C, respectively), which are slightly higher than the estimations corresponding to a burial in a 30 °C/km-gradient (72 and 86 °C respectively). The mean value calculated from the Guéthary flysch measures (70 °C) is far below the estimated temperature (119 °C). The reliability of these data is weakened by the numerous fluid inclusions removed because of reequilibration (whose abundance is discussed later) and the small number of successful measures, i.e. without experimental reequilibration. We assume that the fluid inclusions from the Guéthary flysch samples are unreliable in terms of trapping conditions, as important post-trapping transformations seem to have occurred. By contrast, the Haizabía and Socóa results are very coherent, as emphasized by their pre-folding fracture samples that provide very close average T_h values (78 and 74 °C, respectively).

5. Clay mineralogy

5.1. Methods

The <2 µm fraction of 20 marl samples was prepared as 1 mg/cm² sedimented slides, after a 30-min-ultrasonic disaggregation, as recommended by Kisch (1991) and considering the remarks of Krumm and Buggisch (1991). Three oriented slides were prepared for each sample: the first one was directly analysed after air-dried (AD) sedimentation, the second after ethylene-glycol (EG) saturation during 12 h, and the third one after heating (H) at 550 °C during 2 h. The X-ray diffraction data were collected with an Inel G3000 goniometer using CuKα radiation at the XRD department of the GET laboratory. The samples were scanned at 30 kV and 40 mA with a step-width of 0.03°2θ during 2 s and patterns were digitally treated with PeakFit (AISN Software) for baseline subtraction. The full-width at half-maximum (FWHM) of illite 10 Å peak was measured from the N- and EG-patterns of each sample and the surface area of the (001) peak of each clay mineral was measured from the EG-pattern of each sample. The systematic error is estimated at less than 5% for these two measurements.

5.2. Results

Patterns obtained from the <2 µm fraction of the marl samples allow for the identification of several clay minerals (Fig. 12). The presence of Ca-rich smectite is evidenced by a broad peak centred at 15 Å on the N-patterns, which shifts to 17 Å on the EG-patterns and to 10 Å on the H-patterns. The presence of illite is revealed by a thin peak at 10 Å on the N-, EG- and H-patterns. The thin peak at 7 Å on the N- and EG-patterns, absent on the H-pattern, is ascribed to kaolinite. The relative proportion of these minerals varies significantly throughout the series (Fig. 13). There is a significant drop in kaolinite proportion towards the bottom of the series from a 55% smectites/23% illite/22% kaolinite assemblage at the top to a 89% smectites/10% illite/1% kaolinite assemblage at the bottom of the series.

The illite crystallinity (FWHM of 10 Å peak) values decrease towards the bottom of the series while dispersion of the data strongly increases (Fig. 14). Therefore, both dispersions of the illite crystallinity and of the fluid salinity data increase with depth. As the significance of this trend and the relevance of a possible common cause need to be supported by further results, they are only discussed shortly in this article.

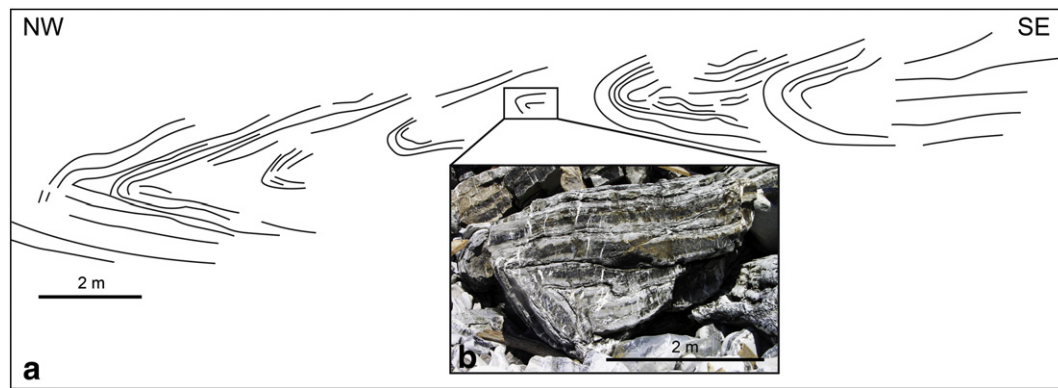


Fig. 7. Recumbent fold of Senix. (a) The very tight interlimb angle of the Senix fold (Fig. 1, outcrop 1), whose axial plane is almost horizontal, has led to the complete dislocation of the hinges along the successive layers. (b) Some of the hinges exhibit vertical veins that cross cut both limbs of the isoclinal fold, as being formed at the end of the folding process.

6. Discussions

6.1. Origin of the fluids salinity

Aqueous solutions of sodium and calcium chlorides have been revealed in the fluids trapped in our calcite samples. Chlorides are documented in more than 95% of sedimentary fluids whereas Na^+ and Ca^{2+} are the most widespread cations in basins where fluid salinity is lower than 40% (Hanor, 1994). The salinity values measured in our samples (0.3 to 23.0‰) are also consistent with referenced data at equivalent depths of burial (<5 km) in sedimentary basins (Hanor, 1979; Ranganathan and Hanor, 1988). Nevertheless, the salinity of water incorporated during deposition is not sufficient to explain such values, and further processes are required.

Evaporite-rich layers may have provided a supply source of chlorides from the lower part of the series (Razin, pers. comm.), as documented by Dorobek (1989). Triassic evaporite deposits are documented in most of the Aquitaine Basin (Kmiecluck and Stevaux, 1971) and have been re-sedimented in the Cretaceous to lower Tertiary series (Henry and Zolnai, 1971). Injected as lubricating horizons along the reverse faults on both sides of the Pyrenees (Sans et al., 1996; Zolnai, 1971), they can be observed along the basal thrust near Bidart, close to the Guéthary flysch outcrops (Fig. 1). Therefore, the high salinity measured in certain samples may be inferred to mainly- NaCl brines produced by dissolution of the evaporites (McManus and Hanor, 1988), as chlorides may have been preferentially solved rather than sulphates (Posey and Kyle, 1988).

Alternatively, a brine-forming process known as membrane filtration or reverse osmosis has been suggested to account for the high-salinity brines found in basins devoid of evaporite (Graf et al., 1966). It involves a hydraulically-driven fluid flow across semi-permeable shale or clay beds (Graf, 1982), which are abundant in the Socoa marly-calcareous series. Thus, considering an upward flow, such a mechanism might have concentrated fluids in the layers underlying the Socoa flysch, which would be consistent with the high salinity of the fluids trapped in the Guéthary flysch. Nonetheless, values lower than the seawater salinity, as found in the Haizabia and Socoa flysch samples, suggest that a meteoric water supply and/or the in situ production of H_2O during diagenesis may have occurred (Hanor, 1994).

6.2. Origin of the clay mineralogy

Clay mineralogy in marl layers results from sedimentary processes (Bjørlykke, 1998) responsible for the primary sediment composition (alimentation and erosion of the source platform, transportation and deposition) and from diagenesis (Alcalá et al., 2012, and references therein). During these early stages, and during deformation

and exhumation, fluid circulation induces a constant reequilibration of the physical and chemical conditions, which results in fluids–mineral reactions (Wilson, 1994). A brief discussion of these potential mechanisms is proposed here to explain some of the mineralogical features found in the marl layers of the Saint-Jean-de-Luz flyschs.

6.2.1. Sedimentary processes

The sedimentary material of the Saint-Jean-de-Luz basin has been deposited under homogenous conditions but the detrital input from the source platform might have changed over time (Razin, 1989). First, the turbidites of the Guéthary flysch were supplied by a bioclastic sedimentation shelf whereas the younger turbiditic systems of Haizabia were connected to marly-calcareous shelves. Such an increasing terrigenous supply is not consistent with the upward increase in the kaolinite content as observed throughout the series. However, Bjørlykke and Aagaard (1992) pointed out the relationships between climate and the kaolinite content in sediments. Following this assumption, Francis and Frakes (1993) suggested that the Turonian and the Coniacian were relatively dry periods, which may be consistent with a low-kaolinite production and the low-kaolinite content of the marls from the Guéthary flysch (lower Coniacian). On the contrary, the Santonian and the early Campanian would have been more humid, in good agreement with a high-kaolinite production and the higher proportion of kaolinite found in the marls of the Socoa and Haizabia flyschs (upper Coniacian–middle Campanian).

6.2.2. Diagenetic processes

In flysch basins where kaolinite is authigenic (Stewart, 1986), low-kaolinite content is due to weak meteoric circulation (Saigal et al., 1992). Such weak meteoric circulation might be responsible for the kaolinite-poor composition of the Guéthary flysch marls, provided that kaolinite is of authigenic origin, which still needs to be demonstrated. Considering important meteoric circulation in the upper formations, the relatively high-kaolinite content in their marl layers would be consistent with the trapping of fluids whose salinity is lower than seawater composition in the corresponding fractures (Hanor, 1994). Besides, several diagenetic reactions lead to the formation of illite from other clay minerals by the means of a K^+ supply (Bjørlykke et al., 1995). Although their kinetics must be very limited at temperatures lower than 150 °C, diagenetically-driven variations in their controlling factors might be partly accountable for variations in the illite content.

The temperature estimated for the deepest formation (119 °C) is consistent with the stability of the clay assemblage; smectites are even observed in sedimentary series at yet higher temperatures. An experimental study of smectites in presence of sodi-calcic fluids, which are characteristic of sedimentary basins, showed that smectites remain

Table 1
Microthermometry data.

Outcrop	Sample	Fracture/vein	Fluid inclusion	Reequilibration	Size (μm)	First melting temperature (°C)	Final ice melting temperature (°C)	Homogenization temperature (°C)	Salinity (wt.% NaCl eq.)
1	RT-11-12	Limb stretching fracture	Pseudo-secondary	Intact	10 × 7	−45.5	−11.5	–	15.5
	RT-11-12	Limb stretching fracture	Pseudo-secondary	Intact	8 × 4	−41.8	−10.6	–	14.6
	RT-11-12	Limb stretching fracture	Pseudo-secondary	Intact	8 × 4	−41.8	−7.4	–	10.9
	RT-11-12	Limb stretching fracture	Pseudo-secondary	Intact	8 × 4	−41.8	−7.4	–	10.9
	RT-11-12	Limb stretching fracture	Primary	Necking down	35 × 15	−50.4	−7.9	–	11.5
	RT-11-12	Limb stretching fracture	Pseudo-secondary	Stretching	7 × 4.5	−41.5	−14.0	–	17.8
	RT-11-12	Limb stretching fracture	Primary	Stretching	90 × 70	−41.1	−14.1	–	17.8
	RT-11-13	Limb stretching fracture	Primary	Necking down	8.5 × 7	−49.4	−20.3	–	22.6
	RT-11-13	Limb stretching fracture	Primary	Necking down	8 × 6	−49.0	−18.9	–	21.6
	RT-11-13	Limb stretching fracture	Primary	Necking down	40 × 8	−41.2	−14.6	–	18.3
	RT-11-13	Limb stretching fracture	Primary	Stretching	70 × 24	−51.1	−20.3	–	22.6
	RT-11-13	Limb stretching fracture	Primary	Stretching	29 × 27	−48.7	−19.9	–	22.3
	RT-11-13	Limb stretching fracture	Primary	Stretching	23 × 13	−48.7	−15.7	–	19.2
2	RT-11-14	Micro-normal fault	Primary	Intact	13.5 × 8.5	−44.7	−6.7	56	10.1
	RT-11-15	Limb stretching fracture	Primary	Intact	6.5 × 6	–	−8.7	–	12.6
	RT-11-15	Limb stretching fracture	Primary	Necking down	16.5 × 10.5	−50.4	−12.2	–	16.1
	RT-11-15	Limb stretching fracture	Primary	Necking down	27 × 22	−46.5	−5.9	–	9.1
	RT-11-15	Limb stretching fracture	Primary	Necking down	18 × 13	−31.7	−8.6	–	12.3
	RT-11-16	Limb stretching fracture	Primary	Intact	34 × 2	−49.7	−16.0	58	19.4
3	RT-11-16	Limb stretching fracture	Primary	Intact	34 × 2	−54.6	−16.0	60	19.4
	RT-11-19	Late fracture	Primary	Stretching	10 × 3	−36.2	−19.0	–	21.7
	RT-11-19	Late fracture	Primary	Stretching	13 × 11.5	−47.4	−12.9	–	16.8
	RT-11-19	Late fracture	Primary	Intact	31.5 × 6	−42.3	−18.1	–	21.0
	RT-11-19	Late fracture	Primary	Stretching	10 × 7	−34.4	−20.9	–	23.0
	RT-11-19	Late fracture	Primary	Intact	5 × 2.5	−49.4	−18.7	139	21.5
	RT-11-19	Late fracture	Primary	Intact	18 × 11	−46.5	−18.4	–	21.3
	RT-11-19	Late fracture	Pseudo-secondary	Intact	5 × 1	−39.6	−17.9	–	20.9
	RT-11-19	Late fracture	Pseudo-secondary	Intact	5 × 1	−39.6	−17.9	–	20.9
	RT-11-20	Late fracture	Primary	Necking down	26 × 5	−51.0	−12.4	–	16.3
4	RT-11-20	Late fracture	Primary	Necking down	25 × 11	−47.8	−8.3	–	12.0
	RT-11-20	Late fracture	Pseudo-secondary	Necking down	5.5 × 5	–	−11.8	–	15.7
	RT-11-20	Late fracture	Primary	Necking down	50 × 30	−48.4	−10.6	–	14.6
	RT-11-20	Late fracture	Pseudo-secondary	Necking down	5.5 × 5	−31.9	−11.7	–	15.6
	RT-11-20	Late fracture	Pseudo-secondary	Necking down	5.5 × 5	−46.7	−11.7	–	15.6
	RT-11-20	Late fracture	Primary	Necking down	40 × 15.5	−47.8	−10.3	–	14.3
	RT-11-20	Late fracture	Primary	Stretching	12 × 10	−50.7	−11.2	–	15.1
	RT-11-20	Late fracture	Primary	Intact	7 × 4	−44.7	−11.8	59	15.7
	RT-11-20	Late fracture	Primary	Intact	6.5 × 2	–	−11.8	59	15.7
	RT-11-20	Late fracture	Primary	Intact	21 × 6	−52.4	−17.2	69	20.4
	RT-11-20	Late fracture	Pseudo-secondary	Intact	8 × 3.5	–	−11.9	78	15.9
	RT-11-21	Late fracture	Pseudo-secondary	Necking down	7 × 3.5	−43.7	−16.0	–	19.5
	RT-11-21	Late fracture	Pseudo-secondary	Necking down	7.5 × 3.5	–	−16.0	–	19.5
	RT-11-21	Late fracture	Primary	Necking down	150 × 7.5	−45.7	−16.9	–	20.2
	RT-11-21	Late fracture	Primary	Necking down	50 × 25	−49.8	−16.9	–	20.2
	RT-11-21	Late fracture	Pseudo-secondary	Intact	4.5 × 2	–	−16.0	42	19.5
	RT-11-21	Late fracture	Pseudo-secondary	Intact	22 × 4.5	−43.7	−16.0	48	19.5
5	RT-11-01	Limb stretching fracture	Primary	Stretching	13 × 33	−36.6	−4.1	–	6.6
	RT-11-01	Limb stretching fracture	Pseudo-secondary	Intact	4.5 × 1.5	−19.9	−1.2	86	2.0
	RT-11-01	Limb stretching fracture	Pseudo-secondary	Intact	4.5 × 1.5	−19.9	−1.2	95	2.0
	RT-11-01	Limb stretching fracture	Primary	Intact	8 × 3	−39.6	−3.3	97	5.5
	RT-11-01	Limb stretching fracture	Primary	Intact	6 × 4	–	−1.7	115	2.8
	RT-11-01	Limb stretching fracture	Primary	Stretching	10.5 × 8.5	−20.4	−1.4	–	2.3
	RT-11-01	Limb stretching fracture	Primary	Intact	7 × 4.5	−20.5	−3.1	124	5.2
	RT-11-01	Limb stretching fracture	Primary	Intact	10 × 7	–	−1.7	127	2.8
	RT-11-05	Diagonal vein (D)	Primary	Necking down	3.5 × 1	–	−0.9	–	1.5
	RT-11-08	Extradors fracture	Primary	Intact	17 × 8	−50.9	−8.1	61	11.8
6	RT-11-35	Diagonal vein (D)	Primary	Intact	10 × 7	−25.8	−3.6	65	5.9
	RT-11-35	Diagonal vein (D)	Pseudo-secondary	Intact	7.5 × 3.5	−39.6	−3.1	74	5.2
	RT-11-35	Diagonal vein (D)	Pseudo-secondary	Intact	1.5 × 1.5	–	−0.2	74	0.3
	RT-11-35	Diagonal vein (D)	Primary	Intact	10 × 8	–	−0.2	–	0.3
	RT-11-35	Diagonal vein (D)	Primary	Stretching	14 × 7	–	−6.0	–	9.2
	RT-11-36	Diagonal vein (D)	Primary	Intact	8.5 × 5.5	−40.6	−1.7	80	2.8
7	RT-11-22	Late fracture	Pseudo-secondary	Necking down	17 × 4.5	–	−0.2	–	0.3
	RT-11-22	Late fracture	Pseudo-secondary	Intact	5 × 4	–	−0.2	107	0.3
	RT-11-22	Late fracture	Pseudo-secondary	Intact	5 × 4	–	−0.2	108	0.3
	RT-11-22	Late fracture	Pseudo-secondary	Intact	5 × 4	–	−0.2	115	0.3
	RT-11-22	Late fracture	Pseudo-secondary	Intact	5 × 3	–	−0.2	147	0.3
	RT-11-22	Late fracture	Primary	Stretching	48.5 × 14	−35.6	−1.7	–	2.8
	RT-11-22	Late fracture	Primary	Intact	56 × 17	–	−0.7	–	1.2
	RT-11-22	Late fracture	Primary	Intact	21 × 13	−39.6	−2.6	–	4.4
	RT-11-22	Late fracture	Pseudo-secondary	Intact	5 × 3	–	−0.2	151	0.3
	RT-11-22	Late fracture	Primary	Necking down	24 × 12	–	−1.1	–	1.9
	RT-11-22	Late fracture	Pseudo-secondary	Intact	7 × 3	−34.6	−2.2	–	3.6

(continued on next page)

Table 1 (continued)

Outcrop	Sample	Fracture/vein	Fluid inclusion	Reequilibration	Size (μm)	First melting temperature (°C)	Final ice melting temperature (°C)	Homogenization temperature (°C)	Salinity (wt.% NaCl eq.)
9	RT-11-27	Diagonal fracture (S)	Primary	Intact	9 × 6.5	–	–0.2	61	0.3
	RT-11-27	Diagonal fracture (S)	Primary	Stretching	9 × 6	–	–0.2		0.3
	RT-11-27	Diagonal fracture (S)	Primary	Intact	20 × 12.5	–	–0.2	80	0.3
	RT-11-28	Diagonal fracture (S)	Primary	Stretching	148 × 6	–50.4	–1.0		1.7
	RT-11-28	Diagonal fracture (S)	Pseudo-secondary	Intact	17 × 12.5	–35.6	–0.4	69	0.7
	RT-11-28	Diagonal fracture (S)	Pseudo-secondary	Intact	9 × 7	–35.6	–0.5	74	0.8
	RT-11-28	Diagonal fracture (S)	Primary	Intact	10 × 6.5	–39.6	–0.2	80	0.3
	RT-11-28	Diagonal fracture (S)	Primary	Intact	10 × 6.5	–39.6	–0.2	80	0.3
	RT-11-28	Diagonal fracture (S)	Primary	Intact	10 × 6.5	–39.6	–0.2	–	0.3
	RT-11-28	Diagonal fracture (S)	Primary	Intact	10 × 6.5	–39.6	–0.2	84	0.3
	RT-11-28	Diagonal fracture (S)	Primary	Intact	12 × 10	–50.3	–2.2	87	3.6
	RT-11-28	Diagonal fracture (S)	Primary	Intact	10 × 5	–39.6	–1.7	124	2.8
	RT-11-28	Diagonal fracture (S)	Primary	Stretching	14 × 9.5	–39.6	–0.2		0.3
	RT-11-28	Diagonal fracture (S)	Primary	Intact	16 × 12	–47.9	–1.5	–	2.5
10	RT-11-30	Diagonal vein (D)	Primary	Necking down	36 × 18	–35.6	–1.4		2.3
	RT-11-30	Diagonal vein (D)	Primary	Intact	60 × 11	–40.6	–0.7	–	1.2
	RT-11-30	Diagonal vein (D)	Primary	Intact	77 × 18	–46.5	–1.2	–	2.0
	RT-11-30	Diagonal vein (D)	Primary	Intact	26 × 16	–21.8	–1.0	–	1.7
	RT-11-30	Diagonal vein (D)	Primary	Stretching	65 × 10	–28.4	–0.6		1.0
	RT-11-30	Diagonal vein (D)	Primary	Stretching	18 × 6	–	–0.2		0.3
	RT-11-32	Diagonal vein (D)	Primary	Intact	30 × 15	–35.6	–1.1	80	1.9
	RT-11-32	Diagonal vein (D)	Primary	Intact	36 × 4	–	–1.2	92	2.0
	RT-11-32	Diagonal vein (D)	Primary	Intact	25 × 23	–	–1.2	–	2.0
	RT-11-33	Interlayers vein	Pseudo-secondary	Intact	6 × 3.5	–	–1.4	53	2.3
	RT-11-33	Diagonal vein (D)	Pseudo-secondary	Intact	6 × 3.5	–	–1.1	53	1.9
	RT-11-33	Diagonal vein (D)	Pseudo-secondary	Intact	17 × 3.5	–	–0.2	74	0.3

Outcrops numbered as in Fig. 1: 1–4: Guéthary flysch samples; 5–7: Socoa flysch samples; 9–10: Haizabab flysch samples.

stable up to 300 °C (Guillaume et al., 2004). In addition, recrystallization of illite seems unlikely since the illite crystallinity measured in the most deeply buried samples would have been homogenized.

6.2.3. Fluid circulation

The similar increasing dispersion of illite crystallinity data and salinity data from fluid inclusions gives rise to the question of their possible

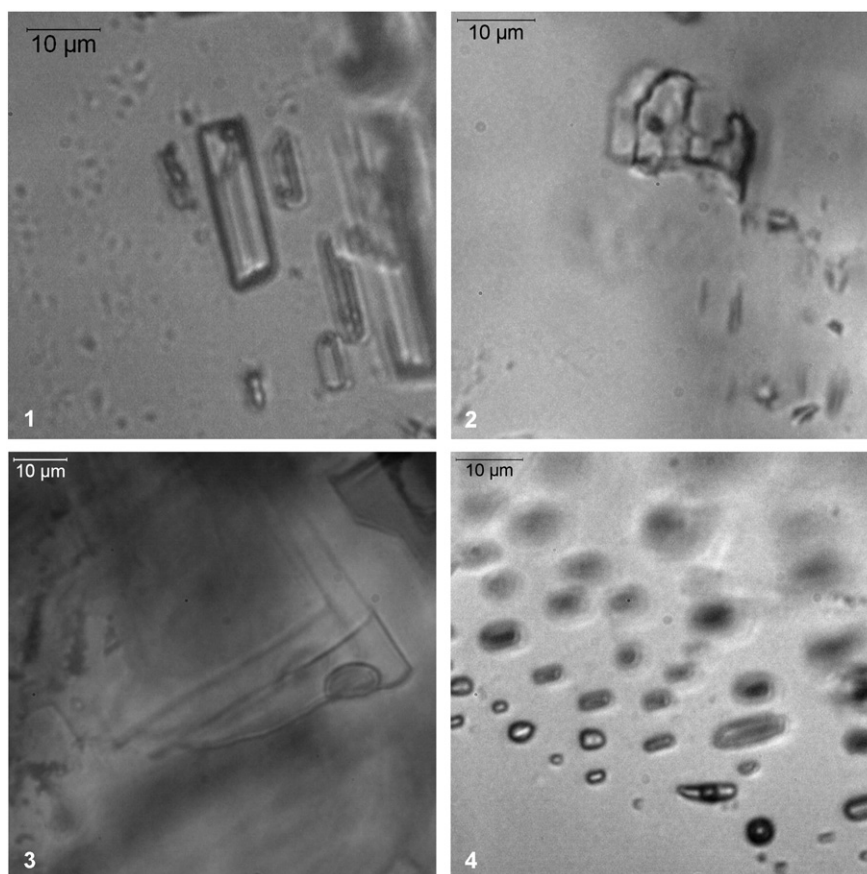


Fig. 8. Photomicrographs of fluid inclusions. (1) Primary fluid inclusion in a late-fracture sample from the Guéthary flysch. (2) Primary inclusion with clues of possible necking down, yet irregularly shaped, in a syn-folding fracture sample from the Guéthary flysch. (3) Primary inclusion showing evidence of stretching in a late-fracture sample from the Socoa flysch; note the size of the vapour bubble. (4) Planar distribution of a pseudo-secondary fluid inclusions group in a sample from an early-folding fracture in the Haizabab flysch.

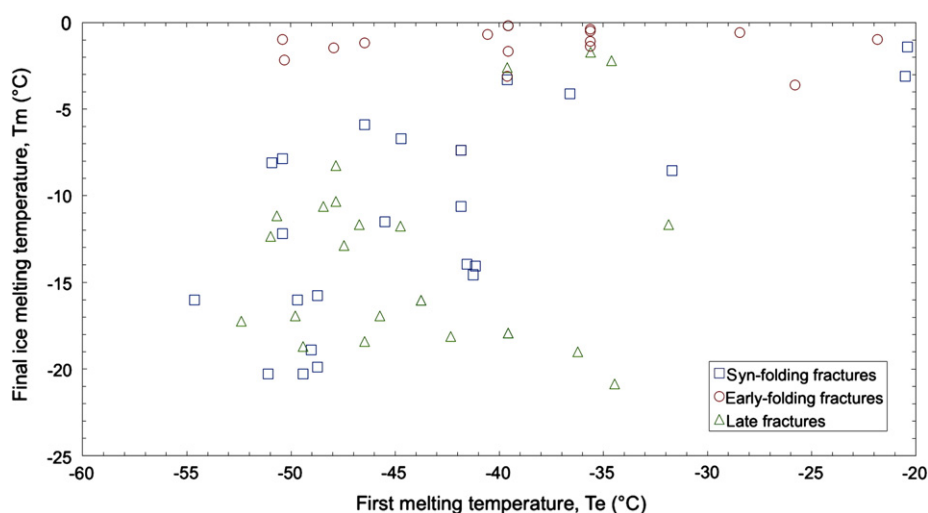


Fig. 9. Te/Tm plot. Data for all the fluid inclusions from the three kinds of fractures; the $\text{H}_2\text{O}-\text{CaCl}_2-\text{NaCl}$ ternary system is evidenced by the values restricted below the corresponding eutectic temperature (55.0°C); note the increasing dispersion towards the highest final ice melting temperatures (T_m), corresponding to the lowest salinities.

common origin. Theoretically, the more fluids circulate (which induces changes in the controlling factors of clay mineral transformations, Essene and Peacor, 1995), the more the fluid–rock equilibrium, and consequently the kinetics of dissolution reactions, are likely to change. In the Guéthary flysch, and to lesser extent in the Socoa flysch, heterogeneous fluid–rock conditions might be regarded as a reasonable consequence of increased fluid circulation, which would result in the higher dispersion of the salinity data and of the illite crystallinity data. Such interaction between fluids trapped in calcite veins and carbonate and phyllosilicate minerals has already been considered for instance by Mazzarini et al. (2010). As suggested for kaolinite, the detrital origin of illite cannot be ruled out and might be an alternative explanation for the dispersion of the crystallinity data.

6.3. Influence of the deformation

Stratification of fluids prior to deformation of the host rocks is documented in modern and ancient sedimentary basins (e.g. Evans and

Battles, 1999; Fischer et al., 2009). In most cases (Deming, 1994, and references therein), salinity of basin fluids increases with depth, as observed in the Saint-Jean-de-Luz basin. Such gradient can give an insight into the processes and salinity sources that are potentially involved. Although mass transfers may have occurred at local scale (Canole et al., 1997, for the Socoa flysch), they could not account for the overall volume of calcite exposed in veins and fractures at series scale, which would have required a 10^4 to 10^5 higher volume of mineralizing fluid (Bjørlykke, 1994).

The vertical salinity gradient suggests an important upward flow of fluid that is a priori not consistent with both the horizontal anisotropic porosity (Du Rouchet, 1981) of alternating calcarenites and marl layers, and the low values (<2%) of bulk porosity measured in the Socoa flysch (Odonne and Massonnat, 1992). However, Secor (1965) demonstrated that the vertical permeability can be enhanced by extensional fractures, i.e. formed perpendicularly to the bedding, such as the fractures related to localized extensional stresses in the Guéthary flysch (Fig. 3.2), and to a lesser extent in the Socoa flysch. Since the highest

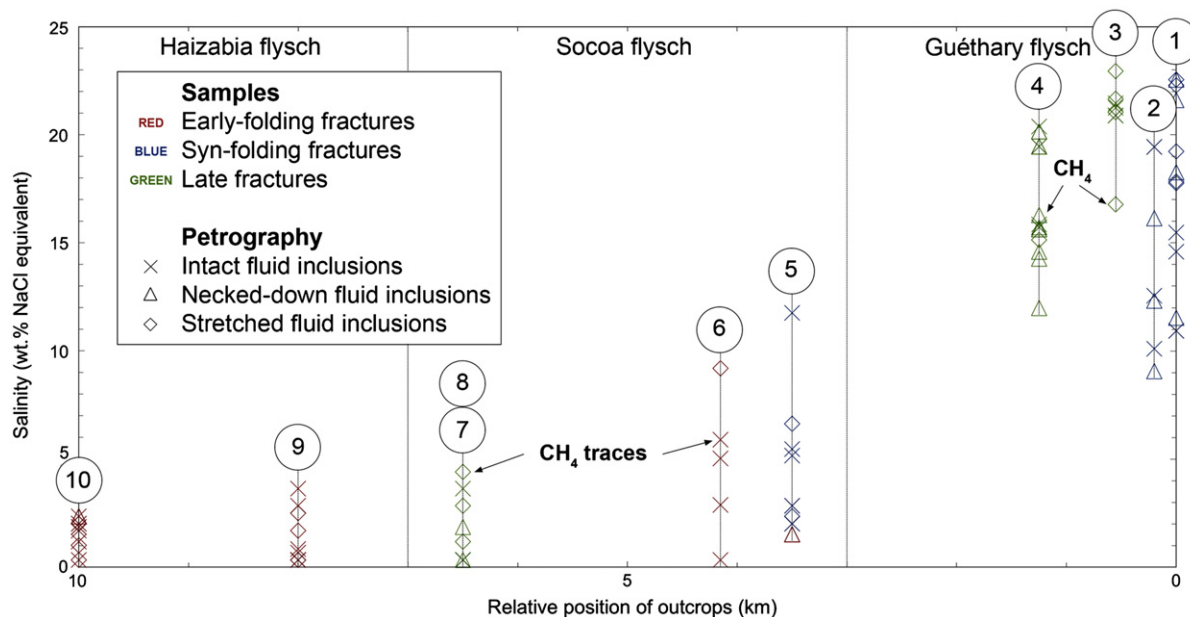


Fig. 10. Synthesis of salinity data. Converted T_m measures for the three kinds of fractures. Both the salinity values and the dispersion of data increase northwards, towards the bottom of the series; the mean values for Haizabia, Socoa and Guéthary flyschs are 1.3%, 3.0% and 17.3%, respectively. Detections of CH_4 by Raman spectroscopy are mentioned with the corresponding fluid inclusions.

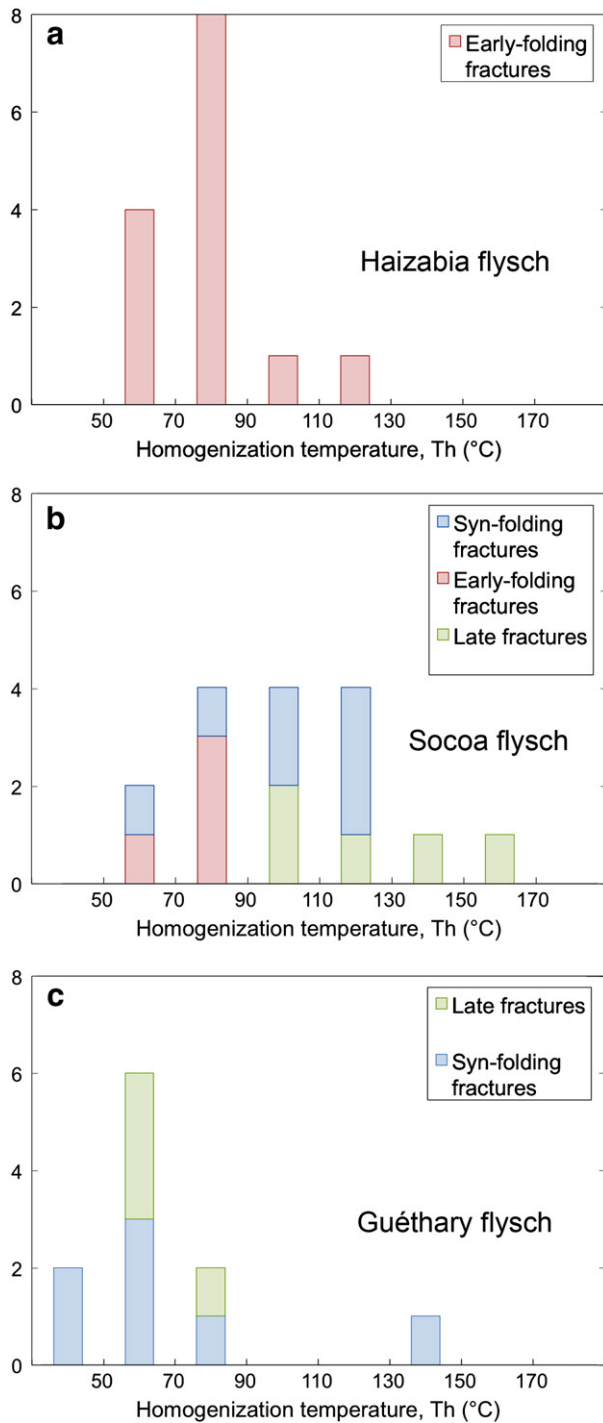


Fig. 11. Synthesis of homogenization temperatures data. Histograms of the Th measures for the three kinds of fractures in the Haizabia (a), Socoa (b) and Guéthary (c) flyschs. Unlike for the Guéthary flysch (70 °C), the mean values for the Socoa (102 °C) and Haizabia (78 °C) flyschs agree with temperatures estimated from the burial of the corresponding formations (119 °C, 86 °C and 72 °C, respectively).

water fluxes in sedimentary basins are expected to occur along open fractures (Bjørlykke, 1994), fracturing related to the late vertical veins has probably contributed to such fluid circulation. As a consequence, significant vertical circulation has been favoured throughout the lower part of the series, where the late fractures are abundant, as opposed to the shallowest formation of Haizabia, where they are not observed.

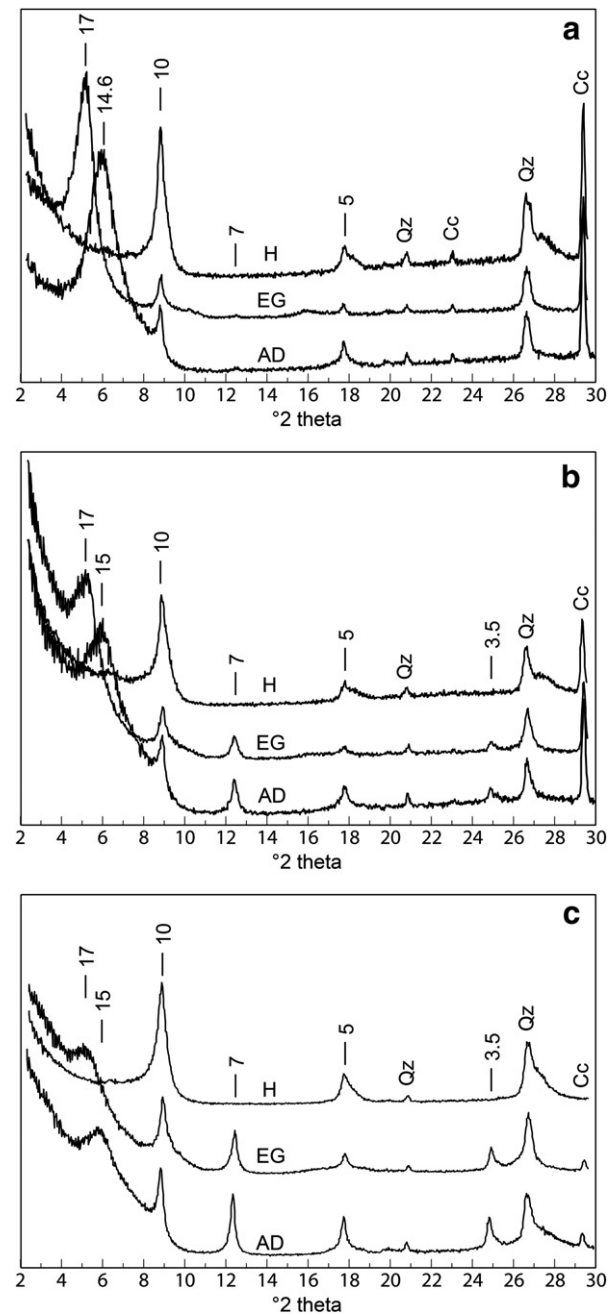


Fig. 12. Clay mineralogy of marl layers. X-ray diffraction patterns of air dried (AD-), ethylene-glycol saturated (EG-) and heated (H) oriented slides of 3 marl samples from the Guéthary (a), the Socoa (b) and the Haizabia (c) flyschs. Explanation for the identification of the clay minerals is given in the text. The 7 Å, 10 Å, and 17 Å peaks of the EG-patterns are used for the quantification of kaolinite, illite and smectite, respectively. Cc: calcite; Qz: quartz.

During tectonic processes and exhumation, reequilibration of fluid inclusions may occur by the means of mechanisms modifying the density of the trapped fluid and/or its composition (Goldstein, 1986, for sedimentary rocks). When the internal pressure sufficiently exceeds the confining pressure, the edges of the fluid inclusions suffer stresses that can increase their volume (Bodnar and Bethke, 1984) and thus reduce the fluid density. This results in abnormally high Th compared to intact fluid inclusions. Since an external fluid supply is unlikely to occur in such overpressured conditions, salinity of the trapped fluid is expected to remain unchanged. Compositional reequilibration would rather be related to propagation of cracks, which benefits from the

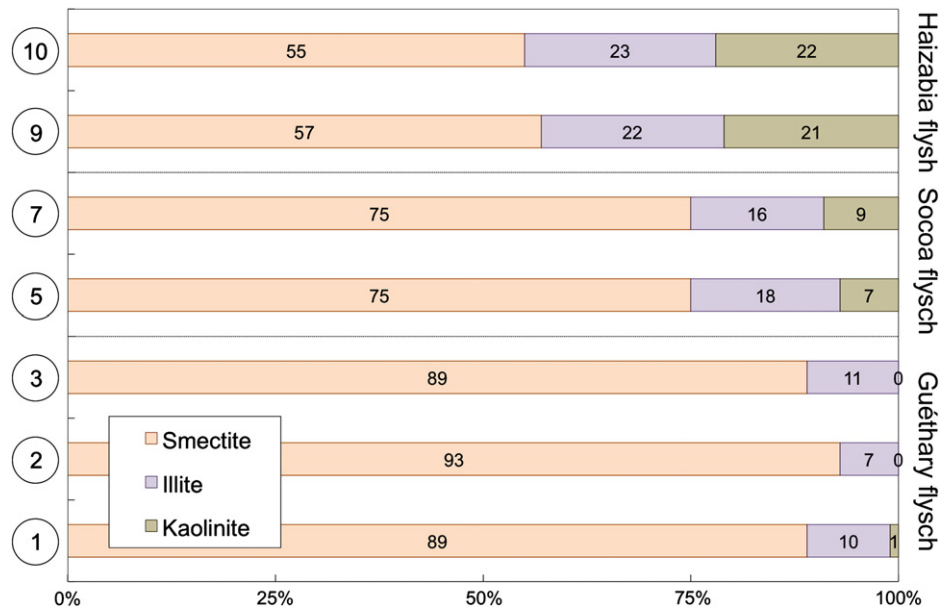


Fig. 13. Proportion of clay minerals. Mean relative abundance values are given for each sampled outcrops; the kaolinite proportion decreases towards the bottom of the series. The sampled outcrops are numbered as in Fig. 1.

easy cleavage of calcite, and might connect the fluid inclusions with porosity fluid whose composition may differ from the trapped fluid. Several groups of fluid inclusions whose petrographic study suggested propagated cracks turned out to be homogeneous in terms of salinity. Therefore, that possibility has been rejected from the possible explanations of the dispersion of the salinity values, yet it might have occurred sparsely. Hydrofracturing can also be responsible for external fluid connection (Burruss and Hollister's, 1979), depending on rheological and morphological characteristics of fluid inclusions and host minerals and on the P/T conditions (Goldstein, 1986). As healing of the fractures and shrinking of the fluid inclusions by necking-down have obscured the required petrographic clues, hydrofracturing is hardly identifiable, particularly for the samples from the Guéthary flysch. Because of numerous questionable or clearly reequilibrated fluid inclusions in the Guéthary flysch samples, further investigation might focus on this formation, even though the microthermometric study of such deformed calcite samples would still remain problematic.

6.4. Synthetic model of fluid circulation

The downward increase of the trapped fluid salinity is regarded as a consequence of a mainly-horizontal fluid circulation and of interactions with rocks within the porous clastic layers. A meteoric contribution is suggested for these fluids as it might explain the low-salinity measured in the fluid inclusions of the upper formations, and has impacted the clay mineralogy as discussed above. In addition, the mixing of those fluids with an upward supply of high-salinity fluids seems to be required to explain the large dispersion of data for the lower part of the series (Fig. 10). Therefore, the proposed model suggests that the strong deformation of the lower formations has enhanced the fracture permeability, noteworthy by increasing the volume of late and extensional vertical fractures. The resulting increased fluid circulation through these fractures would have allowed more deformation by strain softening, which in turn would have had a feedback on the fluid circulation, by controlling the permeability of the deformed rock (Wang and Park,

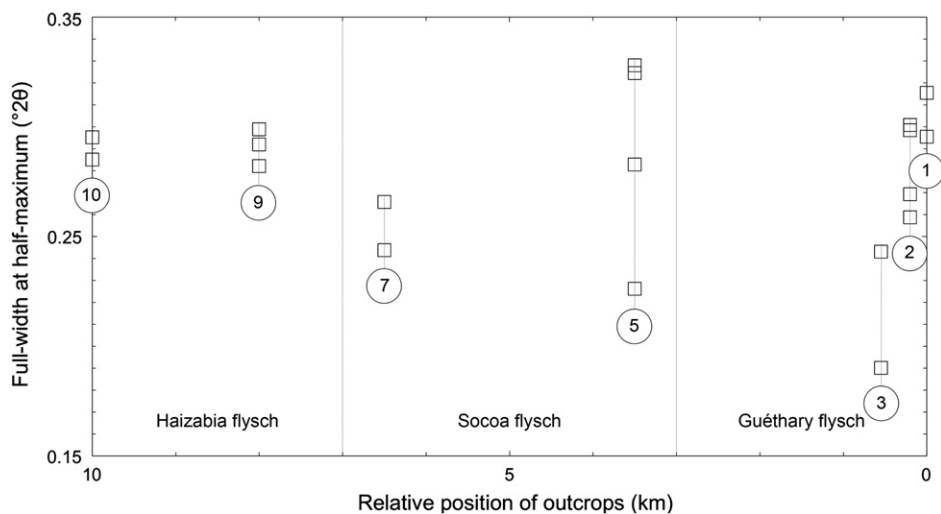


Fig. 14. Illite crystallinity. Evolution in the series of the full-width at half-maximum of illite 10 Å peak. The dispersion of data increases towards the bottom of the series; the mean values for Haizabzia, Socoa and Guéthary flyschs are 0.2907°2θ, 0.2785°2θ and 0.2714°2θ respectively. The sampled outcrops are numbered as in Fig. 1.

2002). Possible enrichments of the fluids through interactions with evaporite-rich layers down the series and with marly layers in the Socoa flysch (membrane filtration) are accounted in this model

(Fig. 15). Such mixing of low-salinity fluids with brines related to evaporite-rich layers has been documented in another flysch basin by Montomoli et al. (2001).

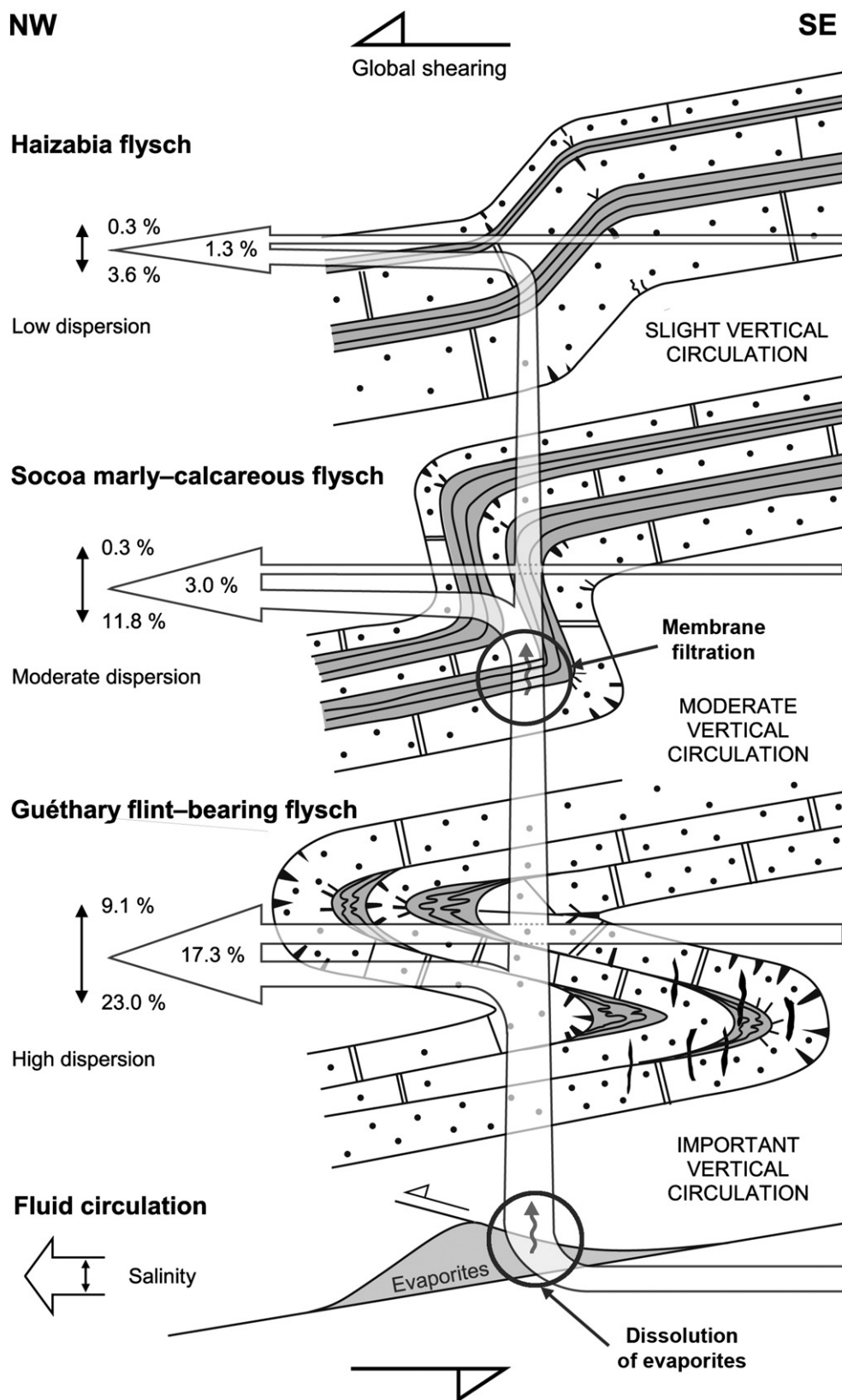


Fig. 15. Main features of the synthetic model of fluid circulation. Mixing of horizontal and vertical fluid circulations is represented by the arrows whose thickness corresponds to salinity; the double arrows show the dispersion of salinity data with minimum and maximum values mentioned for each formation. The vertical scale and both the deformational and the salinity gradient are arbitrarily fitted.

7. Conclusions

The proposed approach has given rise to a preliminary model that constrains the physico-chemical conditions associated with the folding and fracturing of the Cretaceous flyschs on the French Basque coast. Microthermometric investigation of fluid inclusions in calcite crystals from the various veins and fractures reveals an increasing salinity towards the bottom of the series (0.3–23.0 wt.% NaCl equivalent), in agreement with referenced data for sedimentary basin fluids. Increasing dispersion of the salinity data, from the Haizabia flysch (0.3–1.3 wt.% NaCl eq.) to the Guéthary flysch (9.1–23.0 wt.% NaCl eq.) suggests the mixing of mainly-horizontal fluid circulation, potentially related to meteoric water, with an upward supply of high-salinity fluids from evaporite-rich underlying layers. Abundant cross-cutting fractures observed in the lower part of the series, i.e. in the Guéthary and Socoa flyschs, are regarded as a possible explanation for such vertical circulation.

Analysis of clay mineralogy by X-ray diffraction reveals that the kaolinite content drops from 22% to 1% between the top and the bottom of the series. This trend might be related to the transition to a more humid climate proceeded between Turonian and Campanian, but diagenesis and especially fluids–mineral reactions resulting from fluid circulation have yielded several clues of their potential impact.

Microthermometric measurement of T_h from the Guéthary flysch samples provides a mean trapping temperature (70 °C) that is inconsistent with temperature estimated for the burial (119 °C). This data is regarded as poorly reliable given the numerous reequilibrated fluid inclusions observed in these samples and confirms the limitations of P/T reconstruction by microthermometry from strongly deformed samples of diagenetic mineral such as calcite. Nonetheless, the trapping temperatures measured in the Haizabia flysch (78 °C) and the Socoa flysch (102 °C) are in good agreement with the burial estimations. As the geothermal gradient of 30 °C/km seems to be a fair assessment, the depth of burial could not have exceeded a few kilometres depth, i.e. less than 5 km, which is consistent with the clay mineral assemblage found in the marl layers of the series.

To sum up, the evolution of fold geometry, occurrence of fractures, fluid composition and trapping conditions from the southern to the northern French Basque coast, i.e. downward the series of Cretaceous flysch, are a great illustration of the interdependence of deformational gradient and fluid circulation in sedimentary formations.

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References

- Alcalá, F.J., López-Galindo, A., Martín-Martín, M., 2012. Clay mineralogy as a tool for integrated sequence stratigraphic and palaeogeographic reconstructions: Late Oligocene–Early Aquitanian western internal South Iberian Margin, Spain. *Geological Journal*. <http://dx.doi.org/10.1002/gj.2451>.
- Allmendinger, R.W., Cardozo, N., Fisher, D., 2012. *Structural Geology Algorithms: Vectors and Tensors in Structural Geology*. Cambridge University Press.
- Anastasio, D.J., Bebout, G.E., Holl, J.E., 2004. Extra-basinal fluid infiltration, mass transfer, and volume strain during folding: insights from the Idaho–Montana thrust belt. *American Journal of Science* 304, 333–369.
- Bjørlykke, K., 1993. Fluid flow in sedimentary basins. *Sedimentary Geology* 86, 137–158.
- Bjørlykke, K., 1994. Fluid flow processes and diagenesis in sedimentary basins. In: Parnell, J. (Ed.), *Geofluids: Origin, Migration and Evolution of Fluids in Sedimentary Basins*. J. Geol. Soc. London Spec. Publ., 78, pp. 127–140.
- Bjørlykke, K., 1998. Clay mineral diagenesis in sedimentary basins—a key to the prediction of rock properties. Examples from the North Sea Basin. *Clay Minerals* 33, 15–34.
- Bjørlykke, K., Aagaard, P., 1992. Clay minerals in north sea sandstones. In: Houseknecht, D.W., Pittman, E.D. (Eds.), *Origin, Diagenesis, and Petrophysics of Clay Minerals*. SEPM Special Publication, 47, pp. 65–80.
- Bjørlykke, K., Aagaard, P., Egeberg, P.K., Simmons, S.P., 1995. Geochemical constraints from formation water analyses from the North Sea and Gulf Coast Basin on quartz, feldspar and illite precipitation in reservoir rocks. In: Cubitt, J.M., England, W.A. (Eds.), *The Geochemistry of Reservoir*. J. Geol. Soc. London Spec. Publ., 86, pp. 33–50.
- Bodnar, R.J., 2003a. Interpretation of data from aqueous-electrolyte fluid inclusions. In: Samson, I., Anderson, A., Marshall, D. (Eds.), *Fluid inclusions: Analysis and Interpretation*. Mineral. Assoc. Can., Short Course Ser., 32, pp. 81–100.
- Bodnar, R.J., 2003b. Introduction to fluid inclusions. In: Samson, I., Anderson, A., Marshall, D. (Eds.), *Fluid inclusions: Analysis and Interpretation*. Mineral. Assoc. Can., Short Course Ser., 32, pp. 1–8.
- Bodnar, R.J., 2003c. Re-equilibration of fluid inclusions. In: Samson, I., Anderson, A., Marshall, D. (Eds.), *Fluid inclusions: Analysis and Interpretation*. Mineral. Assoc. Can., Short Course Ser., 32, pp. 213–230.
- Bodnar, R.J., Bethke, P.M., 1984. Systematics of stretching of fluid inclusions. I. Fluorite and sphalerite at 1 atmosphere confining pressure. *Economic Geology* 79, 141–161.
- Bodou, P., 1972. Etude de la tectonique de la Côte Basque entre Hendaye et Bidart (Pyrénées Atlantiques). D.E.S., Strasbourg (68 pp.).
- Burke, E.A.J., 2001. Raman microspectrometry of fluid inclusions. *Lithos* 55, 139–158.
- Burruss, R.C., Hollister, L.S., 1979. Evidence from fluid inclusions for a paleogeothermal gradient at the geothermal test well sites, Los Alamos, New Mexico. *Journal of Volcanology and Geothermal Research* 5, 163–177.
- Canole, P., Odonne, F., Polve, M., 1997. Heterogeneous strain associated with normal faulting: evidence of mass transfer by pressure solution associated with fault displacement. *Tectonophysics* 283, 129–143.
- Deming, D., 1994. Fluid flow and heat transport in the upper continental crust. In: Parnell, J. (Ed.), *Geofluids: Origin, Migration and Evolution of Fluids in Sedimentary Basins*. J. Geol. Soc. London Spec. Publ., 78, pp. 27–40.
- Diamond, L.W., 2003. Introduction to gas-bearing aqueous fluid inclusions. In: Samson, I., Anderson, A., Marshall, D. (Eds.), *Fluid Inclusions: Analysis and Interpretation*. Mineral. Assoc. Can., Short Course Ser., 32, pp. 101–158.
- Dorobek, S., 1989. Migration of orogenic fluids through the Siluro–Devonian Helderberg Group during late Paleozoic deformation: constraints on fluid sources and implications for thermal histories of sedimentary basins. *Tectonophysics* 159, 25–45.
- Du Rouchet, J., 1981. Stress fields, a key to oil migration. *American Association of Petroleum Geologists Bulletin* 65, 74–85.
- Essene, E.J., Peacor, D.R., 1995. Clay mineral thermometry—a critical perspective. *Clays and Clay Minerals* 43, 540–553.
- Evans, M.A., 2010. Temporal and spatial changes in deformation conditions during the formation of the Central Appalachian fold-and-thrust belt: evidence from joints, vein mineral paragenesis, and fluid inclusions. In: Tollo, R., Bartholomew, M.J., Hibbard, J., Karabinos, P. (Eds.), *From Rodinia to Pangea: the Lithotectonic record of the Appalachian Region*. Geological Society of America Memoir, 206, pp. 477–552.
- Evans, M.A., Battles, D.A., 1999. Fluid inclusion and stable isotope analysis of veins from the central Appalachian Valley and Ridge province: implications for regional synorogenic hydrologic structure and fluid migration. *Geological Society of America Bulletin* 111, 1841–1860.
- Evans, M.A., Bebout, G.E., Brown, C.H., 2012. Changing fluid conditions during folding: an example from the central Appalachians. *Tectonophysics* 576–577, 99–115.
- Feuillée, P., Sigal, J., 1965. Les calcaires de Béohobie (Basses-Pyrénées, France) et Guipuzcoa (Espagne). *Comptes Rendus. Académie des Sciences* 260, 2016–2019.
- Fischer, M.P., Higuera-Diaz, I.C., Evans, M.A., Perry, E.C., Leticariu, L., 2009. Fracture-controlled paleohydrology in a map-scale detachment fold: insights from the analysis of fluid inclusions in calcite and quartz veins. *Journal of Structural Geology* 31, 1490–1510.
- Francis, J.E., Frakes, L.A., 1993. Cretaceous climates. In: Wright, V.P. (Ed.), *Sedimentology Review*, 1. Blackwell Publishing Ltd., Oxford, pp. 17–30.
- Gamond, J.F., 1983. Displacement features associated with fault zones: a comparison between observed examples and experimental models. *Journal of Structural Geology* 5, 33–45.
- Goldstein, R.H., 1986. Reequilibration of fluid inclusions in low-temperature calcium-carbonate cement. *Geology* 14, 792–795.
- Goldstein, R.H., 2001. Fluid inclusions in sedimentary and diagenetic systems. *Lithos* 55, 159–193.
- Goldstein, R.H., 2003. Petrographic analysis of fluid inclusions. In: Samson, I., Anderson, A., Marshall, D. (Eds.), *Fluid inclusions: Analysis and Interpretation*. Mineral. Assoc. Can., Short Course Ser., 32, pp. 9–53.
- Goldstein, R.H., Reynolds, T.J., 1994. Systematics of fluid inclusions in diagenetic minerals. *Soc. Sed. Geol., Short Course SEPM*, 31 (199 pp.).
- Graf, D.L., 1982. Chemical osmosis, reverse chemical osmosis, and the origin of subsurface brines. *Geochimica et Cosmochimica Acta* 46, 1431–1448.
- Graf, D.L., Meents, W.F., Shimp, N.F., 1966. The Origin of Saline Formation Waters, III: Calcium Chloride Waters, 397. Illinois Geological Survey Circular (60 pp.).
- Guillaume, D., Neaman, A., Cathelineau, M., Mosser-Ruck, R., Peiffert, C., Abdelmoula, M., Dubessy, J., Villiéras, F., Michau, N., 2004. Experimental study of the transformation of smectite at 80 °C and 300 °C in the presence of iron oxides. *Clay Minerals* 39, 17–34.
- Hanor, J.S., 1979. The sedimentary genesis of hydrothermal fluids. In: Barnes, H.L. (Ed.), *Geochemistry of Hydrothermal Ore Deposits*, 2nd ed., pp. 137–172.
- Hanor, J.S., 1994. Origin of saline fluids in sedimentary basins. In: Parnell, J. (Ed.), *Geofluids: Origin, Migration and Evolution of Fluids in Sedimentary Basins*. J. Geol. Soc. London Spec. Publ., 78, pp. 151–174.
- Henry, J., Zolnai, G., 1971. Sur le Trias résédimenté dans le Sud-Ouest du Bassin Aquitain. *Bulletin du Centre de Recherches de Pau-SNPA* 5, 389–398.

- Kisch, H.J., 1991. Illite crystallinity: recommendations on sample preparation, X-ray diffraction settings, and interlaboratory samples. *Journal of Metamorphic Geology* 9, 665–670.
- Kmieluck, C., Stevaux, J., 1971. Le bassin triasique d'Aquitaine et ses épisodes évaporitiques. *Bulletin du Centre de Recherches de Pau-SNPA* 5, 345–355.
- Krumm, S., Buggisch, W., 1991. Sample preparation effects on illite crystallinity measurement: grain-size gradation and particle orientation. *Journal of Metamorphic Geology* 9, 671–677.
- Lawler, J.P., Crawford, M.L., 1983. Stretching of fluid inclusions resulting from a low-temperature microthermometric technique. *Economic Geology* 78, 527–529.
- Macpherson, G.L., 1989. Source of lithium and barium in Gulf of Mexico formation waters, USA. In: Miles, D.L. (Ed.), *Water Rock, Interaction*, pp. 453–456.
- Marrett, R., Laubach, S.E., 1997. Diagenetic controls on fracture permeability and sealing. *International Journal of Rock Mechanics and Mining Sciences* 34, 204.e1–204.e11.
- Mathy, B., 1986. Les Flyschs Crétacé supérieur des Pyrénées basques. Age, anatomie, origine du matériel, milieu de dépôt et relations avec l'ouverture du golfe de Gascogne. (Thèse de doctorat) Université de Bourgogne, Dijon (403 pp.).
- Mathy, B., Sigal, J., 1976. Sur l'âge et la nature des calcaires de Béthobie (Pyrénées atlantiques et province Guipuzcoa, Espagne). *Compte Rendu. Sommaire des Séances. Société Géologique de France*, Paris 5, 208–210.
- Mazzarini, F., Isola, I., Ruggieri, G., Boschi, C., 2010. Fluid circulation in the upper brittle crust: thickness distribution, hydraulic transmissivity fluid inclusion and isotopic data of veins hosted in the Oligocene sandstones of the Macigno Formation in southern Tuscany, Italy. *Tectonophysics* 493, 118–138.
- McManus, K.M., Hanor, J.S., 1988. Calcite and iron sulfide cementation of Miocene sediments flanking the West Hackberry salt dome, South-west Louisiana, USA. *Chemical Geology* 74, 99–112.
- Meunier, J.D., 1989. Assessment of low-temperature fluid inclusions in calcite using microthermometry. *Economic Geology* 84, 167–170.
- Montomoli, C., Ruggieri, G., Boiron, M.C., Cathelineau, M., 2001. Pressure fluctuation during uplift of the Northern Apennines (Italy): a fluid inclusions study. *Tectonophysics* 341, 121–139.
- Mulder, T., Razin, P., Faugères, J.C., 2009. Hummocky cross-stratification-like structures in deep-sea turbidites: Upper Cretaceous Basque basins (Western Pyrenees, France). *Sedimentology* 56, 997–1015.
- Mullis, J., Dubessy, J., Poty, B., O'Neil, J., 1994. Fluid regimes during late stages of a continental collision: Physical, chemical, and stable isotope measurements of fluid inclusions in fissure quartz from a geotransverse through the Central Alps, Switzerland. *Geochimica et Cosmochimica Acta* 58, 2239–2267.
- Nelson, R.A., 2001. *Geologic Analysis of Naturally Fractured Reservoirs*. Gulf Professional Publishing (332 pp.).
- Odonne, F., Massonnat, G., 1992. Volume loss and deformation around conjugate fractures: comparison between a natural example and analogue experiments. *Journal of Structural Geology* 14, 963–972.
- Olivet, J.L., 1996. La cinématique de la plaque Ibérique. *Bulletin des Centres de Recherches Exploration-Production Elf-Aquitaine* 20, 131–195.
- Prezbindowski, D.R., Larese, R.E., 1987. Experimental stretching of fluid inclusions in calcite - implications for diagenetic studies. *Geology* 15, 333–336.
- Posey, H., Kyle, R., 1988. Fluid-rock interactions in the salt dome environment: an introduction and review. *Chemical Geology* 74, 1–24.
- Poty, B., Stalder, H.A., 1974. Fluid inclusions studies in quartz from fissures of Western and Central Alps. *Schweizerische Mineralogische und Petrographische Mitteilungen* 54, 717–752.
- Ramsay, J.G., 1974. Development of chevron folds. *Geological Society of America Bulletin* 85, 1741–1754.
- Ranganathan, V., Hanor, J.S., 1988. Density driven ground water flow near salt domes. *Chemical Geology* 74, 173–188.
- Razin, P., 1989. Evolution tectono-sédimentaire alpine des Pyrénées Basques à l'Ouest de la transformante de Pampelune (Province du Labourd). (Thèse de doctorat) Université de Bordeaux III (464 pp.).
- Robert, I., 1979. Etude des mécanismes de plissement d'une série stratifiée, dans une zone de cisaillement. Exemple de la Côte Basque. (Thèse de spécialité) Université scientifique et médicale de Grenoble (254 pp.).
- Roedder, E., 1984. Fluid inclusions. *Reviews in Mineralogy*, 12. Mineral. Soc. America (644 pp.).
- Roure, F., Swennen, R., Schneider, F., Faure, J.L., Ferket, H., Guilhaumou, N., Osadetz, K., Robion, P., Vandeginste, V., 2005. Incidence and importance of tectonics and natural fluid migration on reservoir evolution in foreland fold-and-thrust belts. *Oil and Gas Science and Technology*, 60. Revue de l'IFP 67–106.
- Saigal, G.C., Bjørlykke, K., Larter, S.R., 1992. The effects of oil emplacements on diagenetic processes—examples from the Fulmar Reservoir sandstones, Central North Sea. *AAPG Bulletin* 76, 1024–1033.
- Sanford, W.E., Wood, W.W., 1991. Brine evolution and mineral deposition in hydrologically open evaporate basins. *American Journal of Science* 291, 687–710.
- Sans, M., Sanchez, A.L., Santanach, P., 1996. Internal structure of a detachment horizon in the most external part of the Pyrenean fold and thrust belt (northern Spain). In: Alsop, G.J., Blundell, D.J., Davison, I. (Eds.), *Salt Tectonics*. Geol. Soc. Sp. Pub. Nb., pp. 65–76.
- Secor, D.T., 1965. Role of fluid pressure in jointing. *American Journal of Science* 263, 633–646.
- Shepherd, T.J., Rankin, A.H., Alderton, D.H.M., 1985. *A Practical Guide to Fluid Inclusion Studies*. Chapman & Hall (239 pp.).
- Sibson, R.H., 1994. Crustal stress, faulting and fluid flow. In: Parnell, J. (Ed.), *Geofluids: Origin, Migration and Evolution of Fluids in Sedimentary Basins*. J. Geol. Soc. London Spec. Publ., 78, pp. 69–84.
- Sibuet, J.C., Collette, B., 1991. Triple junctions of Bay of Biscay and North Atlantic: new constraints on the kinematic evolution. *Geology* 19, 522–525.
- Srivastava, S.P., Sibuet, J.C., Cande, S., Roest, W.R., Reid, L.D., 2000. Magnetic evidence for slow seafloor spreading during the formation of the Newfoundland and Iberian margins. *Earth and Planetary Science Letters* 182, 61–76.
- Stewart, D.J., 1986. Diagenesis of the shallow marine Flumar Formation in the Central North Sea. *Clays and Clay Minerals* 21, 537–564.
- Vilasi, N., Malandain, J., Barrier, L., Callot, J.P., Amrouch, K., Guilhaumou, N., Lacombe, O., Muska, K., Roure, F., Swennen, R., 2009. From outcrop and petrographic studies to basin-scale fluid flow modelling: the use of the Albanian natural laboratory for carbonate reservoir characterization. *Tectonophysics* 474, 367–392.
- Wang, J.-A., Park, H.D., 2002. Fluid permeability of sedimentary rocks in a complete stress-strain process. *Engineering Geology* 63, 291–300.
- Wilson, M.D., 1994. Reservoir quality assessments and prediction in clastic rocks. *SEPM Short Course*, 30 (432 pp.).
- Zolnai, G., 1971. Le front nord des Pyrénées occidentales. Histoire structural du golfe de Gascogne. *Technip*, pp. 1–10 (chap. IV).